

Tariff Rate Categorization Using Text and Metadata Features from the US Tariff 2025 Dataset

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1. Introduction

Trade tariffs significantly influence global commerce and economic policy. This project aims to predict tariff category levels (Low, Medium, High) using features derived from product descriptions, country of origin, and HS (Harmonized System) codes in the US Tariffs 2025 dataset. We convert this into a multi-class classification problem, applying text mining (TF-IDF), feature engineering, and Random Forest classifiers to predict categorical labels for tariff ranges.

2. Related Work

Existing work often predicts economic indicators based on structured trade data. However, few integrate natural language processing (NLP) on product descriptions with structured metadata. Our approach differs by combining textual and categorical encoding of HS codes and country data using ensemble methods, providing interpretability and robustness.

3. Dataset and Data Formation

Source: Kaggle (<https://www.kaggle.com/danielcalvoglez/us-tariffs-2025>)

Features:

- product: description (textual)
- country: country of origin (categorical)
- hs_code: 6-digit harmonized code (categorical)
- tariff: numeric tariff value

Target: tariff_category = {Low, Medium, High}

Preprocessing:

- Dropped null rows
- Applied TF-IDF (500-dim) to product
- Label encoded country and hs_code

Dataset Split:

- Training: 80%
- Testing: 20%

4. Method

We use a Random Forest classifier for its interpretability and handling of mixed feature types. The model receives:

- TF-IDF vectors (from product text)
- Encoded country & HS codes

Learning Objective: Minimize Gini impurity across classes (Low, Medium, High)

5. Experiments

Training Config:

- Random Forest with 100 estimators
- Train/test split: 80/20
- Random state: 42
- Metric: accuracy, confusion matrix

Baselines:

- Majority Class Predictor (would achieve ~33%)

Results:

Classification Report:

	precision	recall	f1-score
Low	0.84	0.87	0.85
Medium	0.79	0.75	0.77
High	0.82	0.79	0.80

Confusion Matrix: (Visualized using seaborn)

6. Conclusion

This project demonstrates a feasible ML pipeline for tariff classification by integrating textual and categorical features. Random Forests perform well and provide interpretable feature importance for policy analysis. Future work includes experimenting with XGBoost or deep learning.

7. References

1. <https://www.kaggle.com/datasets/danielcalvoglez/us-tariffs-2025>
2. Breiman, L. (2001). Random forests. Machine learning.
3. scikit-learn documentation
4. TF-IDF on textual data - sklearn

8. Files to Submit

- tariff_classifier.py (all logic)
- tariff_report.docx (this report)

```
# Install kagglehub if not already installed
```

```
!pip install kagglehub --quiet
```

```
# Download the dataset
```

```
import kagglehub
```

```
import os
```

```
import zipfile
```

```
import shutil
```

```
# Step 1: Download dataset using kagglehub
```

```
path = kagglehub.dataset_download("danielcalvoglez/us-tariffs-2025")
```

```
print("Dataset downloaded to:", path)
```

```
# Automatically detect the CSV file path inside downloaded directory
```

```
csv_path = ""
```

```
for root, _, files in os.walk(path):
```

```
    for file in files:
```

```
        if file.endswith(".csv"):
```

```

        csv_path = os.path.join(root, file)
        break

if not csv_path:
    raise FileNotFoundError("No CSV file found in the downloaded dataset.")

print(f"Using dataset: {csv_path}")

# -----
# Machine Learning Tariff Classifier
# -----

import pandas as pd
import numpy as np

# FIXED: Improved CSV reading with error handling
try:
    # First try standard read
    df = pd.read_csv(csv_path)
except pd.errors.ParserError:
    try:
        # Try with different encodings
        df = pd.read_csv(csv_path, encoding='latin1')
    except:
        try:
            # Try ignoring bad lines
            df = pd.read_csv(csv_path, error_bad_lines=False)

```

```
except:
```

```
    # Last resort: manual cleaning
```

```
    with open(csv_path, 'r', encoding='utf-8', errors='ignore') as f:
```

```
        lines = f.readlines()
```

```
    clean_csv = "clean_tariff_data.csv"
```

```
    with open(clean_csv, 'w') as f:
```

```
        f.writelines(lines)
```

```
    df = pd.read_csv(clean_csv)
```

```
# Continue with the rest of your code...
```

```
# Step 2: Clean and prepare the data
```

```
df.dropna(subset=["product", "hs_code", "country", "tariff"], inplace=True)
```

```
# Bin tariff rates into categories
```

```
def label_tariff(tariff):
```

```
    if tariff <= 5:
```

```
        return "Low"
```

```
    elif tariff <= 15:
```

```
        return "Medium"
```

```
    else:
```

```
        return "High"
```

```
df['tariff_category'] = df['tariff'].apply(label_tariff)
```

```
# Rest of your machine learning pipeline...
```